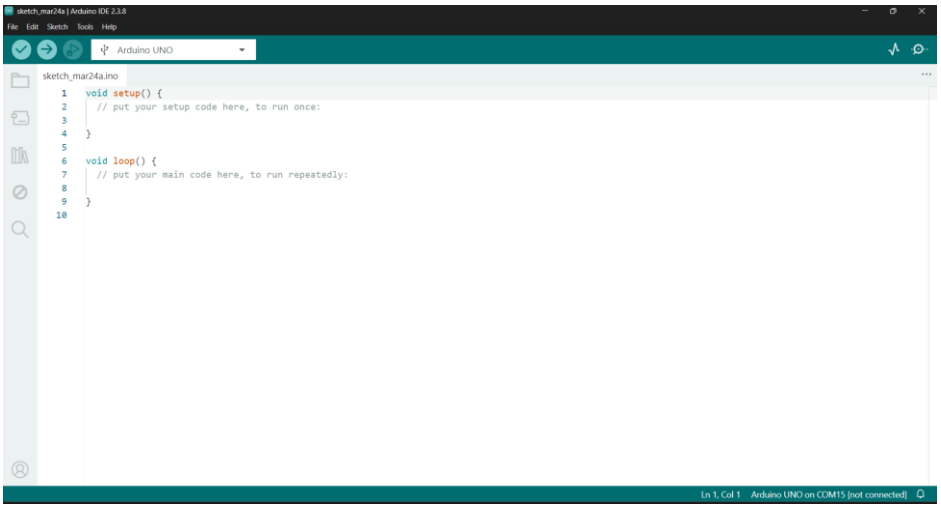
EXERCISE 1: ENVIRONMENT SETUP

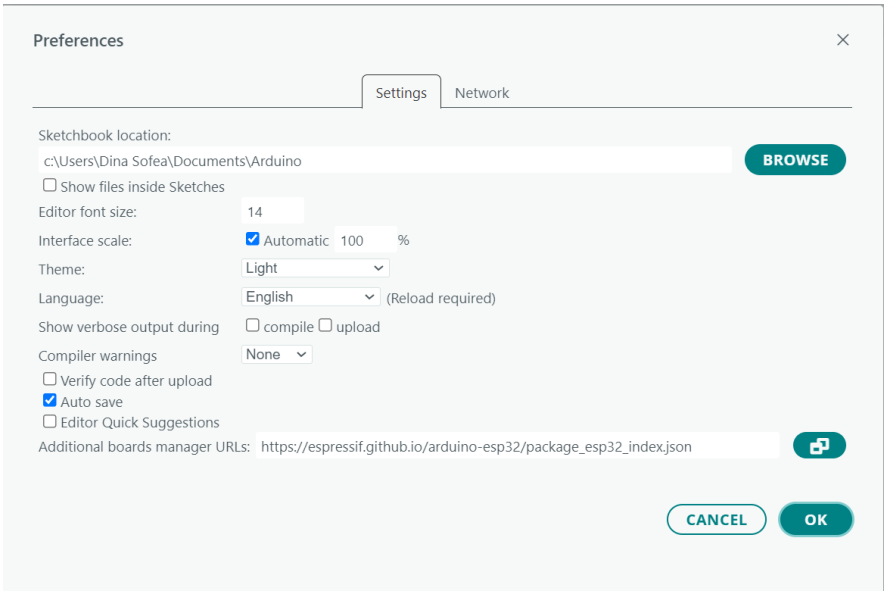
PART 1: CORE LOCAL SOFTWARE TOOLCHAINS

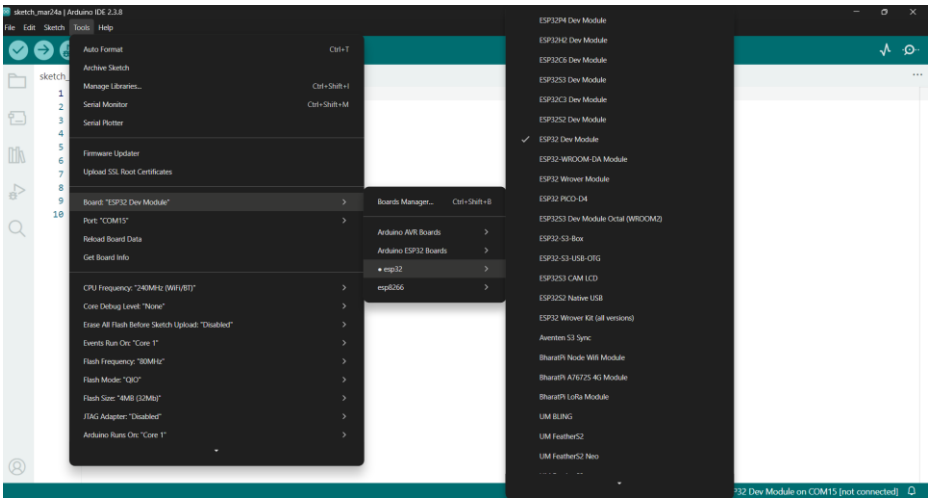
1. Embedded IDE & C/C++ Toolchain

1. **Step 1: Download and install the Arduino IDE.**



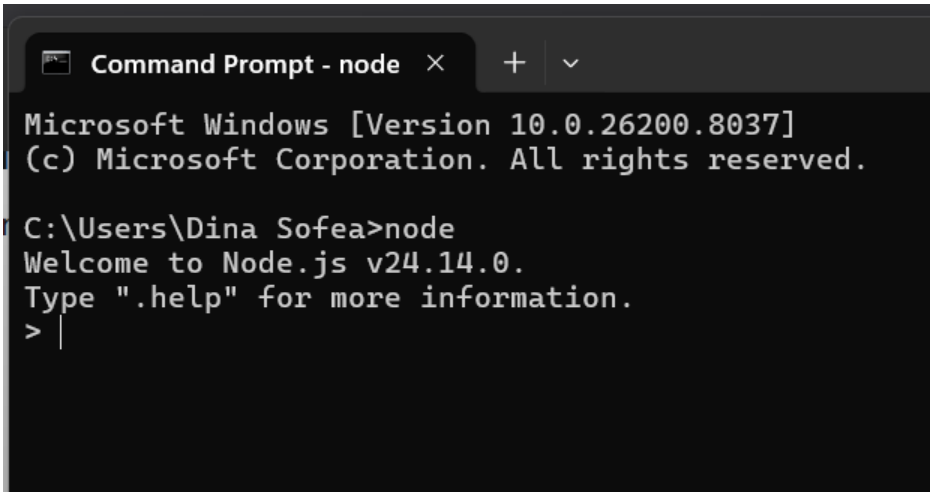
2. **Step 2: Open the extension manager and install the necessary ESP32 board support packages.**





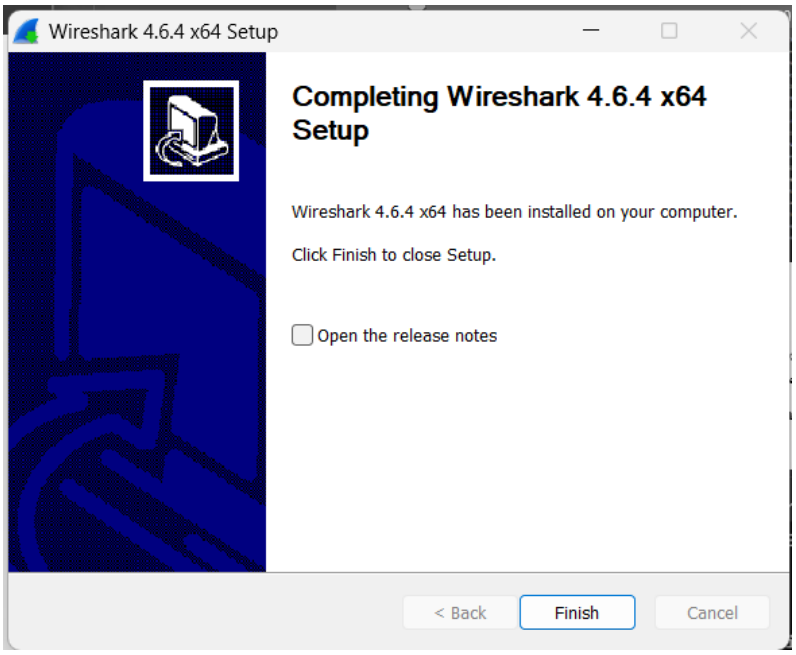
2. NodeJS and NPM

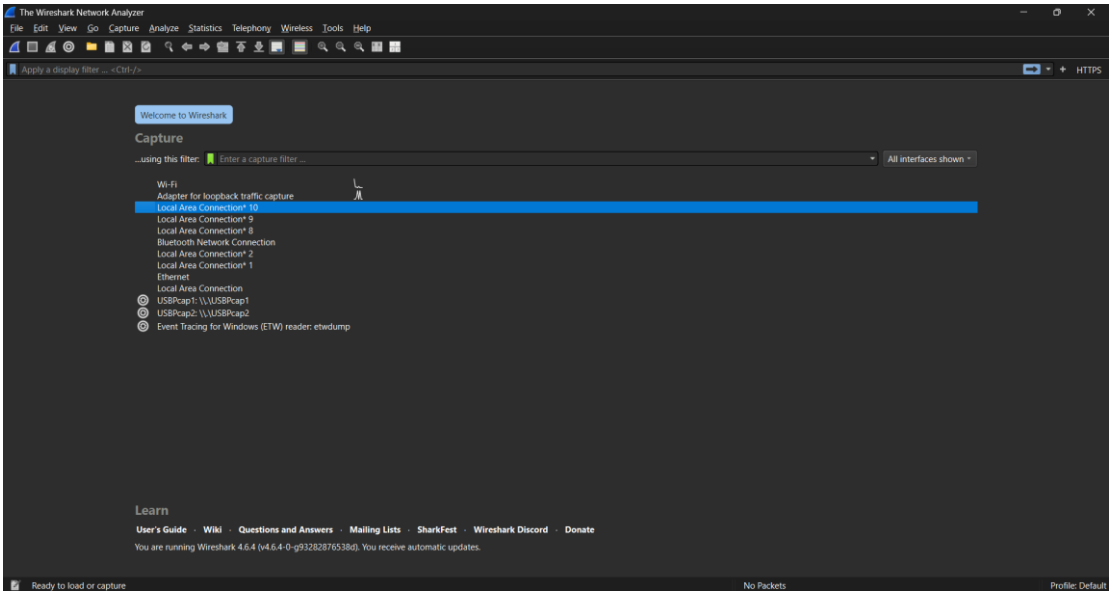
1. **Step 1: Download and install from NodeJs website.**
2. **Step 2: Open the Command Prompt and type "node".**
3. **Step 3: Press CTRL+C to terminate the session.**



3. Wireshark

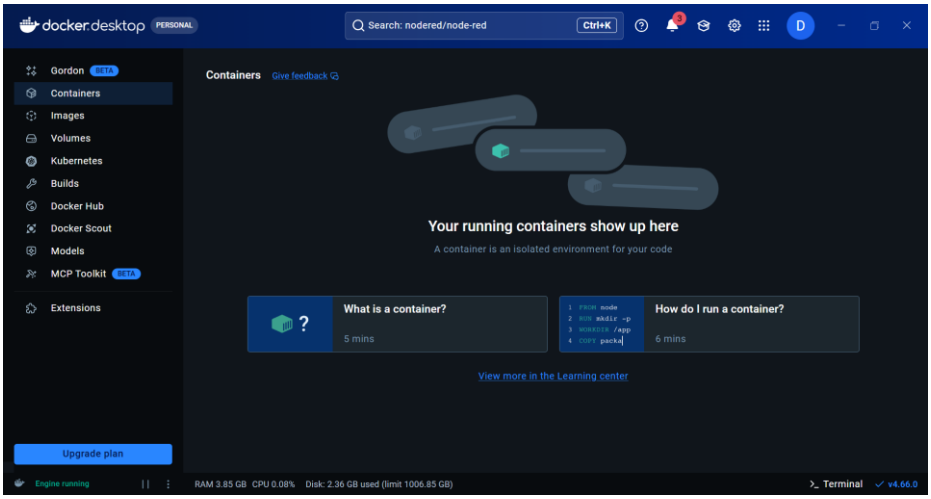
1. **Step 1: Download Wireshark from the official website.**
2. **Step 2: Complete the installation, ensuring you also install the required packet capture libraries (like Npcap for Windows) when prompted by the installer.**


Wireshark 4.6.4 x64 Setup
Completing Wireshark 4.6.4 x64 Setup
Wireshark 4.6.4 x64 has been installed on your computer.
Click Finish to close Setup.
Open the release notes
< Back
Finish
Cancel


The Wireshark Network Analyzer
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
HTTPS
Welcome to Wireshark
Capture
...using this filter:
Enter a capture filter ...
All interfaces shown
Wi-Fi
Adapter for loopback traffic capture
Local Area Connection* 10
Local Area Connection* 9
Local Area Connection* 8
Bluetooth Network Connection
Local Area Connection* 2
Local Area Connection* 1
Ethernet
Local Area Connection
USBPcap1: \\.\USBPcap1
USBPcap2: \\.\USBPcap2
Event Tracing for Windows (ETW) reader: etwdump
Learn
User's Guide · Wiki · Questions and Answers · Mailing Lists · SharkFest · Wireshark Discord · Donate
You are running Wireshark 4.6.4 (v4.6.4-0-g9328287653 8d). You receive automatic updates.
Ready to load or capture
No Packets
Profile: Default

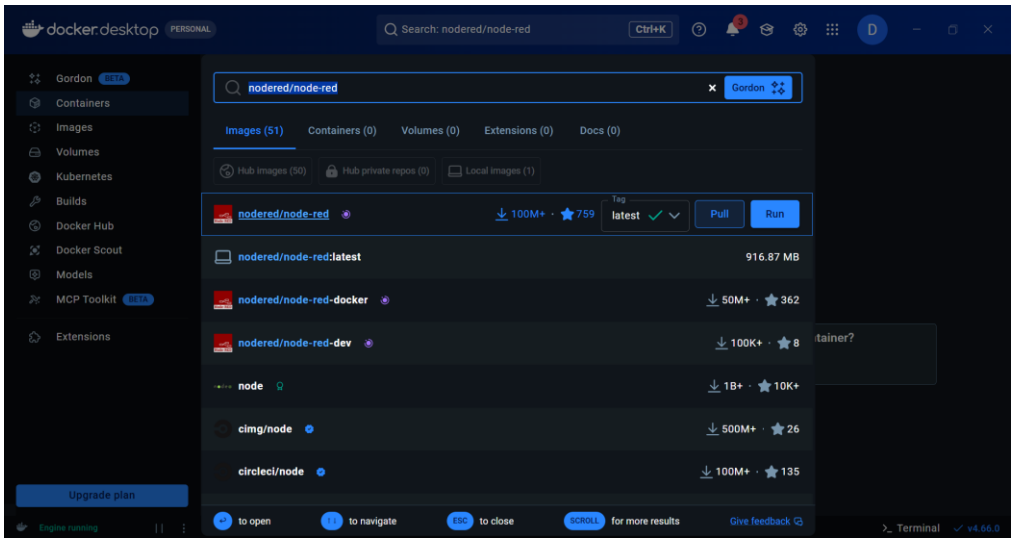
PART 2: CONTRANERIZED LOCAL INFRASTRUCTURE (DOCKER)

1. **Docker Desktop Installation**
    1. **Step 1: Navigate to the Docker website and download Docker Desktop for your operating system.**
    2. **Step 2: Follow the installation prompts and ensure the Docker Engine is running in your system tray.**



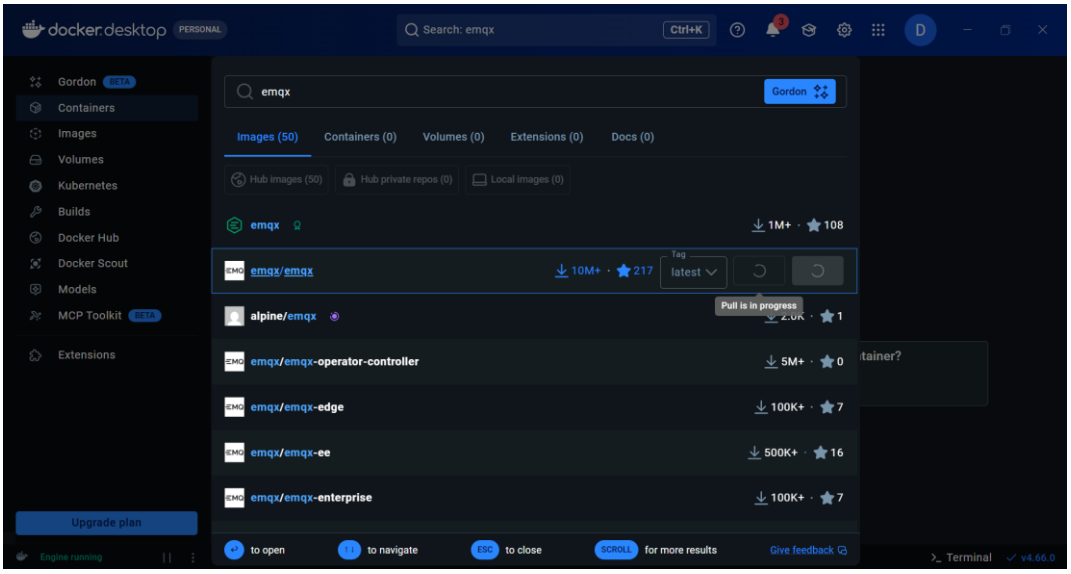
2. Node-RED (Local Middleware)

1. **Step 1: Open Docker Desktop and click on the search bar at the top (which connects directly to Docker Hub).**
2. **Step 2: Search for the official image: nodered/node-red.**
3. **Step 3: Click "Pul" to download the image to your local machine so it is ready for deployment in upcoming labs.**



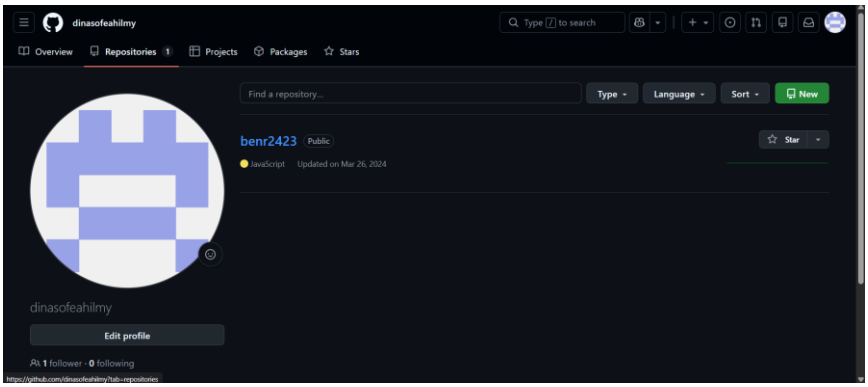
3. Node-RED (Local Middleware)

1. **Step 1: In the Docker Desktop search bar, search for the official image: emqx/emqx.**
2. **Step 2: Click "Pull" to download the broker image.**

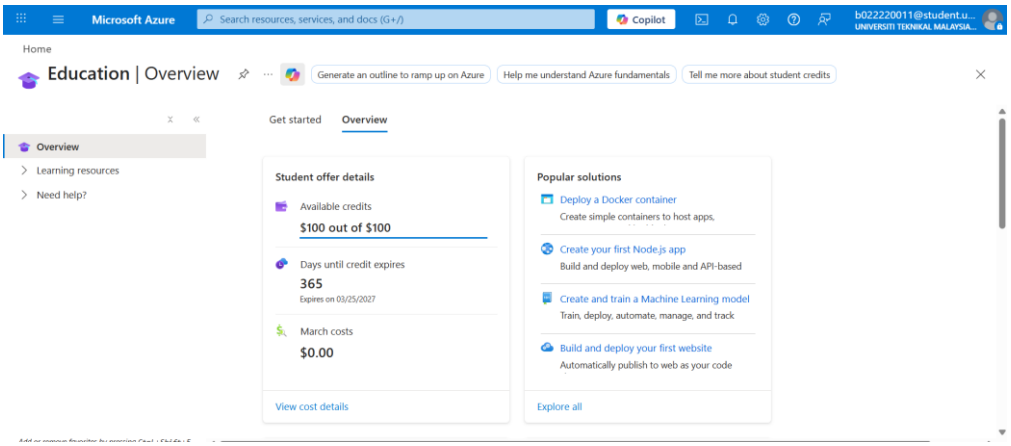
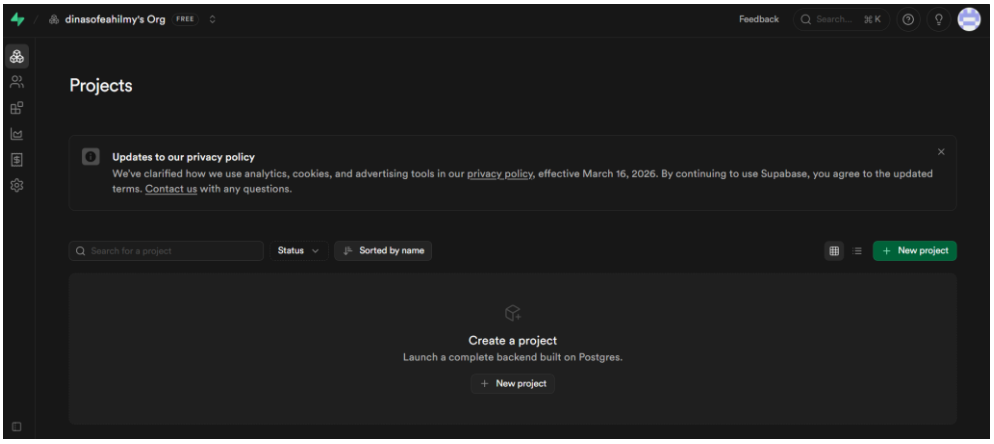


PART 3: CLOUD & DEVELOPER ACCOUNT REGISTRATION

1. Github Account



2. Microsoft Azure for Students



3. Supabase